Leverage Network and Contagion*

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Abstract

Using unique account-level data that track hundreds of thousands of margin investors' leverage ratios, trading activities, and portfolio holdings at a daily frequency, we examine the interaction between funding conditions and asset prices during the recent stock market turmoil in China. We show that idiosyncratic shocks to individual stocks can be amplified and transmitted to other securities with which they share common ownership by levered investors. Further, using a network-based approach, we show that stocks that have more and stronger connections to other stocks through common holdings by levered investors (i.e., that are more central in the leverage network) tend to have lower returns and higher crash risk going forward.

Keywords: margin trading, leverage, contagion

JEL Classification: G11, G23

I. Introduction

Investors can use margin – that is, the ability to lever up their positions by borrowing against the securities they hold – to amplify returns. A well-functioning lending-borrowing market is crucial to the functioning of the financial system. In the Capital Asset Pricing Model, for instance, investors with different risk preferences lend to and borrow from one another to clear both the risk-free and risky asset markets. Just like any other type of short-term financing, however, the benefit of margin trading comes at a significant cost: it makes investors vulnerable to the temporary fluctuations in funding conditions. For example, a levered investor may be forced to liquidate her positions if the value of her portfolio drops temporarily below some pre-negotiated level.

A growing theoretical literature carefully models this two-way interaction between security returns and leverage constraints (e.g., Gromb and Vayanos, 2002; Geanakoplos, 2003; Brunnermeier and Pedersen, 2009). The core idea is that an initial reduction in security price lowers the collateral value, thus making the leverage constraint more binding. This then leads to additional selling by some levered investors and depresses the price further, which could trigger even more selling by levered investors and an even lower price. Such a downward spiral can dramatically amplify the initial adverse shock to security value, the degree to which depends crucially on the characteristics of the margin traders that are holding the security. (A similar mechanism, albeit to a less extent, is also at work with an initial, positive shock to security value.)

This class of models also makes predictions across assets. When faced with pressure to deleverage, investors may indiscriminately downsize all their holdings, even those that have not gone down in value and thus have little to do with the tightening of leverage constraints. This indiscriminate selling pressure could generate a contagion across assets that are connected solely through common holdings by levered investors. In other words, idiosyncratic shocks to one security can be amplified and transmitted to other securities through a latent leverage network structure. In some situations (in the spirit of Acemoglu, Carvalho, Ozdaglar, and Tahbaz-Salehi (2012)), idiosyncratic shocks to individual securities, propagated through the leverage network, can aggregate to and result in systematic price movements.

Despite their obvious importance to academics, investors and regulators, testing the asset pricing implications of margin trading, however, has been empirically challenging, due to the unavailability of account-level leverage data. In this paper, we fill this gap in the literature by taking advantage of unique account-level data from China that track hundreds of thousands of margin investors' borrowing and trading activities at a daily frequency.

Our dataset covers an interesting three-month period – from May to July 2015 – during which the Chinese stock market experienced a rollercoaster ride: the Shanghai Composite Index climbed 15% from the beginning of May to its peak at 5166.35 on June 12th, before crashing nearly 30% by the end of July. Major financial media around the world have linked this incredible boom and bust in the Chinese stock market to the growing popularity of, and subsequent government tightening on, margin trading in China.[1] Indeed, as evident in Figure 1, the aggregate amount of margin borrowing through brokerages and the Shanghai Composite Index moved in lockstep (with a correlation of over 90%) for the period from October 2014 to September 2015. This is consistent with the narrative that the ability to buy stocks on margin fueled the initial stock market boom and the subsequent de-leveraging exacerbated the bust.

Insert Figure 1 about Here

Our data, obtained from a major brokerage firm in China as well as a web-based trading platform that facilitates peer-to-peer lending in the stock market, contain detailed records of individual accounts' leverage ratio, along with their holdings at the end of each trading day in the three-month period. The granularity of our data allows us to directly examine the impact of margin trading on asset prices – specifically, how idiosyncratic shocks to individual firms, transmitted through the nexus of leveraged accounts, can cause a contagion in the equity market and eventually aggregate to systematic movements.

Our preliminary evidence is strongly supportive of the view that margin trading has significant impact on asset prices. First, we show that idiosyncratic shocks in the market can cause contagion across assets when these assets are "connected" through common holdings by margin investors. In particular, the returns of one security can be strongly and positively forecasted by the returns of other securities with which it shares a common margin-investor base.

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[1] For example, "Chinese firms discover margin lending's downside," Wall Street Journal, June 30, 2015; "China's stock market crash: A red flag," Economist, July 7, 2015; "China cracks down on margin lending before markets reopen," Financial Times, July 12, 2015.

This result shows up strongly both in a Fama-MacBeth regression, and easily survives the inclusion of control variables for individual stock's own leverage and other known predictors of stock returns in the cross-section.

Further, using a network-based approach, we show that stocks that are connected to a larger number of other well-connected stocks through common holdings by margin investors – i.e., stocks that are more central to the leverage network – tend to have lower returns and higher crash risk going forward. .

The rest of the paper is organized as follows. Section 2 gives background information about the Chinese stock market and regulations on margin trading. Section 3 discusses our data and screening procedures. Section 4 presents our main empirical results. Finally, Section 5 concludes.

2. Institutional Backgrounds of the Chinese Margin Trading System

The last two decades have witnessed tremendous growth in the Chinese stock market. As of May 2015, the total market capitalization of China's two stock exchanges, Shanghai Stock Exchange (SSE) and Shenzhen Stock Exchange (SZSE), exceeded 10 trillion dollars, second only to the US market. Despite the size of the stock market, margin trading was not popular in China. In fact, before 2010, margin financing was not officially authorized in the Chinese stock market, although it occurred informally on a small scale. The China Securities Regulatory Commission (CSRC) launched a pilot program for margin financing by brokerage firms in March 2010 and margin financing was officially authorized in October 2011. To obtain margin financing from a registered brokerage firm, investors need to have a trading account with that brokerage for at least 18 months,[2] with account value (cash and stock holdings combined) exceeding RMB500,000 (about USD80,000). The initial margin is set at 50% and the maintenance margin is typically 23%. A list of around 900 stocks is chosen by the CSRS as eligible for margin buying and short selling; the list is periodically reassessed and modified.

The aggregate broker-financed margin debt has grown exponentially since its introduction. Starting from mid-2014, it has closely tracked the performance of the Chinese stock market and peaked around RMB 2.26 trillion yuan in June 2015 (see Figure 1). It is about 3% to 4% of the total market capitalization of the stock market in China. This ratio is similar to that found in New York Stock Exchange (NYSE) for example. The crucial difference is that margin traders in China are mostly retail investors who are prone to speculation, rather than sophisticated institutional investors in the US who are well equipped with risk management tools and derivative contracts to hedge their risks. Compared to margin buys, short selling is much more difficult as only six brokerages firms initially allowed their registered account holders to short sell. In addition, derivative contracts on individual stocks are very limited and in general not effective.[3]

Catering to investors' demand for leverage, shadow-finance margin trading has also

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[2] This account-opening requirement was lowered to six months in 2013.

[3] A small subset of Chinese stocks issued put warrants during the past few years. Xiong and Yu (2011) show that such market is highly speculative, and provides hardly any hedging services.

become popular since 2014. These informal financing arrangements come in many shapes and forms, but most of them allow investors to take on even higher leverage when speculating in the stock market. For example, Umbrella Trust is a popular arrangement where a few wealthy investors or a group of smaller investors provide an initial injection of cash, for instance 20 percent of the total trust's value. The remaining 80 percent is funded by selling stakes to other, usually retail investors, in the form of wealth management products promising juicy yields – typically higher than the risk-free rate. As such, the umbrella trust structure implies a leverage ratio of 5. In addition, it also allows small retail investors to bypass the RMB 500,000 minimum threshold required for obtaining margin financing from brokerage firms. The extreme popularity of e-commerce in China made it possible for software companies to develop web-based trading platforms with shadow financing capabilities.[4] Some of these web-based trading platforms allow further split of an umbrella trust, increasing the effective leverage even more. Finally, shadow-financed margin trading allows investors to take levered positions on any stocks, including the ones not on the marginable security list.

Since shadow-financed margin trading falls in an unregulated grey area, there is no official statistic regarding its size and effective leverage ratio. Estimates of its total size from various sources range from RMB 0.8 trillion yuan to 3.7 trillion yuan. It is widely believed that the amount of debt in the shadow financing system is as big as the amount of leverage financed through the formal broker channel. For example, Huatai securities Inc., one of China's leading brokerage firms, estimates that the total margin debt to be around 7.2% of the market capitalization of the Chinese stock market and 19.6% of the market capitalization of its free float.

[4]HOMS, MECRT, and Royal Flush were the three largest OTC-based margin trading platforms in China.

Excessive leverage through the shadow financing system is often blamed for causing the dramatic stock market gyration in 2015. Indeed, in mid-June 2015, CSRC ruled that all online trading platforms must stop providing leverage to their investors. By the end of August, such leverage trading accounts almost disappeared completely from the Chinese market.

3. Data and Preliminary Analysis

Our paper takes advantages of two unique proprietary account-level datasets. The first dataset contains complete equity holdings, cash balance, order submission, and trade execution records of all existing accounts from a leading brokerage firm for the period of May to July of 2015. This dataset contains over 5 million accounts, 95% of which are retail accounts and around 180,000 accounts are eligible for margin trading. For each margin account, the dataset provides the end-of-day debit rate, defined as the account's total value (cash and equity holding) divided by its outstanding debt. The CSRC mandates a minimum debit rate of 1.3, translating to a maintenance margin of about 23% (= (1.3-1)/1.3).

To check the coverage and representativeness of our brokerage-account data, we aggregate the daily trading volume and corresponding RMB amount across all accounts in our data. Our dataset, on average, accounts for 10% - 15% of the total trading reported by the stock exchanges. Similarly, we find that the total amount of debt taken by all margin accounts in our dataset accounts for about 10% of the total outstanding broker-financed margin debt in the market. We also compute the cross-sectional correlations between the trading volume in our dataset and its market-level counterpart and find the correlation to be over 95% on average. Our sample is therefore representative of the market, at least in terms of trading activities.

Our second dataset contains all the trading and holding records for nearly 200,000 accounts from a major Chinese web-based trading platform from January to July 2015. This dataset includes all information as in the brokerage firm dataset, except that it does not provide end-of-day debit ratios. Instead, the dataset provides detailed information on the initial debt and cash flows between the principle-account and the agent-account, with the agent-account directly linked to stock trading activities. We can thus manually back out the end-of-day wealth and debt value for each agent-account. The dataset also provides details about the wealth-debt ratio that will trigger a margin call, which varies across accounts.

Unlike the brokerage account data, the trading-platform account data include mostly shadow-financed margin investors. Since margin traders in this platform will link their agent-accounts to non-margin accounts in brokerage firms, it is possible that there are overlaps between our brokerage firm non-margin accounts and trading-platform agent-accounts.

In addition to the two proprietary account-level datasets, we also acquire intraday level-II data, as well as daily close price, trading volume, stock return and other stock characteristics from WIND database. The level-II data includes details on each order submitted, withdrawn, and executed, and the price at which the order is executed in the Chinese stock market.

3.1. Computing Leverage Ratios

Before we conduct any statistical analyses, it is crucial to define an accurate measure of leverage for each trading account. Following prior literature (For example, Ang et al (2011)), we define the leverage ratio as:

$$Leverage\ Ratio = \frac{Total\ Wealth}{Total\ Wealth\ -\ Total\ Debt} \tag{1}$$

To obtain the daily leverage ratio for the brokerage-financed margin account, we divide the debit ratio by the absolute difference of this debit ratio and 1. For each shadow-financed margin account, we compute its daily leverage ratio using the inferred daily account wealth and debt value. Note that our web-based account data provides detailed information on daily cash inflow and outflow to each agent account. These daily cash flows, combined with the initial margin debt when the account was first opened, allow us to keep track of the margin debt level in the account over time. We assume cash flow exceeding 10% of current margin debt means repaying existing debt or borrowing additional debt.[5] We measure the total wealth of the account by taking the sum of its equity holdings and cash balance. The resulting leverage ratio varies substantially across shadow-financed accounts, in part because different accounts have different initial margin ratios and maintenance margin ratios as these ratios are negotiated between the investor (the borrower) and the intermediary lending firm (the lender). As a result, shadow-financed margin accounts typically have a much higher leverage ratio than brokerage-financed accounts. It is not surprising that the leverage ratio of some shadow-financed account often exceeds 10. In contrast, the maintenance margin ratio of 0.23 in the brokerage-financed margin account translates to a maximum leverage ratio of 4.33 (= 1/0.23).

Insert Figure 2 about Here

Figure 2 plots the weighted average leverage ratios in both the brokerage-financed margin data and the web-based shadow-financed margin data. We use each account's wealth as the

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[5] This 10% benchmark is suggested by the web-based platform experts. We try other cutoff of 15% and 5%, the results in our study do not change qualitatively.

weight in computing the average ratios. We find that, although in general the leverage ratio in the shadow-financed margin accounts is significantly higher than that in the brokerage accounts, the two ratios display a very similar trend during the overlapping period.

Figure 2 shows that the weighted average shadow-financed leverage ratio in general decreases from January to mid-June, 2015. This trend coincides with the run-up of Shanghai composite index during the same period (see Figure 1), suggesting that the decreasing leverage ratio during the first half of 2015 was probably due to the increase in investor wealth rather than active de-leveraging by the investors. Indeed, as evidenced in Figure 1, outstanding margin debt was increasing during the first half of 2015. Figure 2 also shows a sudden and dramatic increase in the leverage ratios of both the brokerage- and shadow-financed accounts starting in the mid of June, followed by a big crash in the first week of July. Consistent with the market performance in Figure 1, the skyrocket of leverage ratio in mid to late June was likely due to the plummet of market valuation, whereas the decrease in leverage ratio thereafter was likely due to de-leveraging activities, either voluntarily by the investors themselves or forced by the lending intermediaries. Figures 1 and 2 show dramatic fluctuations in the leverage constraints from May to July, we therefore focus on this three-month period for our main analysis where we have data from both the brokerage-financed and shadow-financed accounts.

3.2. Sample Summary Statistics

Insert Table 1 about Here

Table 1 presents summary statistics of our sample. During the three-month period from May to July 2015, our brokerage-financed account data sample contains more than 5 million accounts, among which less than 180,000 are margin accounts. Our web-based shadow-financed account data sample contains a little over 200,000 margin accounts. We first compute the outstanding debt and wealth in terms of cash and equity holdings for each account at the daily frequency, and then sum these measures for the subsamples of the brokerage-financed margin accounts, brokerage non-margin accounts, and the shadow-finance margin accounts, respectively. The results indicate that in both margin account subsamples, about half of the wealth is financed with debt. Figure 1 shows that the average total outstanding debt during the period from May to July is about 200 billion Yuan. Our sample therefore covers roughly 6-10% of total margin debts in the Chinese market.

In Panel B of Table 1, we compare some account trading and holding characteristics across the three subsamples of accounts from May to July, 2015. Since the focus of our analysis is margin trading, we do not examine all the 4.5 million non-margin accounts at the brokerage firm. Instead, we randomly select 1 million of them to examine. Among the three account types, margin accounts at the brokerage firm are the most active. On average they trade 126,000 shares, submit 15 orders, and hold 369,000 shares per day. Shadow-financed margin accounts are less active in trading, yet they are typically associated with higher leverage ratios than their counterparts at the brokerage firm. Both types of margin accounts are wealthier, and more active than the brokerage non-margin traders. This result echoes the pattern shown in Figure 1 that outstanding debt roughly coincides with the level of the stock market index in China.

4. Empirical Analyses

4.1. Leverage Ratios and Stock Returns

Numerous studies have document the negative relationship between asset leverage ratios and future returns, especially during market downturns. Current empirical studies mainly test this hypothesis using hedge fund leverage data. In this subsection, we examine the relationship between stock leverage ratios and future returns using the margin trader account data in China. Our analysis centers on the three month period of May to July, 2015.

A. Double Sorts on Leverage Ratios and Market Returns

In Table 2, we examine the effect of stock leverage ratios on stock returns using two approaches. Panel A presents the results from a portfolio double sort. We first sort all trading days during our three month period into terciles based on market returns in the previous trading day. Then on each trading day, we sort all stocks into terciles based on their leverage ratios at the end of the previous trading day. We use Shanghai Composite Index return to measure the market return, and use two different methods to compute the stock leverage ratios. The first method uses only public data from the WIND database. The WIND database provides the outstanding brokerage-financed margin debt value for each stock, together with its market capitalization valued using tradable shares. We then compute the leverage as the ratio between the market capitalization and the absolute difference between the market capitalization and the margin debt. In the second method, we pool all the brokerage-financed and shadow-financed margin accounts. We then use each account's holdings of each stock as the weight to compute the weighted-average leverage ratio for each individual stock. Finally, Panel A reports average daily portfolio returns (both equal-weighted and value-weighted).

Our results reveal different portfolio return patterns across different market states. When the market was going up yesterday, there is no discernable pattern between leverage ratios and stock returns today; however, when the market was collapsing yesterday, then high-leverage portfolio significantly under-performs low-leverage portfolio today. This result is robust to both definitions of leverage ratios.

Panel A of Table 2 clearly shows an asymmetric relationship between a stock's current leverage ratio and its future return. This result supports the notion that stock market collapse triggers de-leveraging by margin traders and generates further downward price pressure especially among high-leverage stocks.

B. Stock Leverages, Stock Returns, and Control Variables

To confirm this relationship, we run cross-sectional regressions in Panel B at individual stock level. Specifically, we regress the stock return on its previous day's leverage ratio and other control variables. As in Panel A, we also split the sample into three sub-samples according to the previous-day market returns and perform the regression for boom-market and bust-market separately. The regression equation is as follows:

$$RET_{i,t} = \alpha_t + \beta_l LEVERAGE_{i,t-1} + \beta_c CONTROL_{i,t-1} + \varepsilon_{i,t} \tag{2}$$

$RET_{i,t}$ is the return of stock i on day t. $LEVERAGE_{i,t}$ is the leverage ratio for stock i at the end of day t-1. We use two different ways to compute the leverage ratio for each stock (using either the public data or our proprietary account-level data). $CONTROL_{i,t}$ contains a vector of control variables of stock *i*: its stock return in *t*-1, as well as market capitalization, average daily turnover ratio, cumulative returns, and idiosyncratic volatility measured in the previous month.

Insert Table 2 about Here

Table 2 presents the regression results. The regression coefficients in the boom and bust market conditions exhibit very different patterns. Consistent with the results from portfolio sorts, in the boom market, the leverage ratio is insignificant. In sharp contrast, in the crash period, leverage ratios are significantly negatively related to future stock returns, consistent with our intuition that, in down markets, high leverage leads to funding constraints and forced sales of assets.

Other control variables also show some effects during down market. For example, stock returns are negatively correlated with lagged cumulative stock returns and lagged stock turnover ratios. Both suggest that the majority traders in the Chinese market during the three months might be contrarian investors. Market capitalization is marginally positively related to future stock returns, suggesting during the down market investors tend to dump small cap stocks. Both coefficients are consistent with the speculative behaviors of margin investors.

4.2. Empirical Analyses of Common Holding through Margin Accounts

In this subsection, we examine the effect of margin trading on the co-movement in stock returns. The idea is that a negative idiosyncratic shock to stock A may lead some investors to de-lever. If these investors sell indiscriminately across all their holdings, this selling pressure could cause a contagion among stocks that are "connected" to stock A through common ownership by levered investors. A similar story, albeit to a less extent, can be told for a positive initial shock – for example, as one's portfolio value increases, he/she may take on more leverage to expand his/her current holdings.

From this subsection onwards, we select only the 800 component stocks of the Zhongzheng 800 index (one of Chinese major indices) as our target stocks. The Zhongzheng index is a portfolio of 800 stocks listed in both SSE and SZSE of various sizes. It should be mentioned that, among these 800 stocks, about 630 were included in the marginable stock list for broker-financed margin trades. However, two facts guarantee that our sample do not introduce systematic bias in selecting stocks. First, all the shadow-financed margin traders are not subject to the margin stock list constraint declared by CSRC. Second, broker-financed margin investors can choose to sell margin stocks and stocks not on the margin list in their portfolios equally likely when facing funding constraints. This fact makes all the stocks not in the margin list effective marginable with similar leverage ratios as long as they are put in the same portfolio with the marginable stocks. In unreported tables, we repeat all the analyses in this paper within only the 630 stocks that are listed on the margin list, our results do not change qualitatively.

A. Forecasting Stock Co-Movement with Common Holding by Margin Investors

We measure common ownership by margin investors in a way similar to Anton and Polk (2014). At the end of each day, we measure common ownership of a pair of stocks as the total value of the two stocks held by all leveraged investors, divided by the total market capitalization of the two stocks. We label this variable "stock connectedness" $MCP_{i,j,t}$:

$$MCP_{i,j,t} = \frac{\sum_{m=1}^{M} (S_{i,t}^{m} P_{i,t} + S_{j,t}^{m} P_{j,t}) * L^{m}{}_{t}}{TS_{i,t} P_{i,t} + TS_{j,t} P_{j,t}} \qquad (3)$$

where $S_{i,t}^{m}$ is the number of shares of stock i held by levered investor *m*, $TS_{i,t}$ the number of tradable shares outstanding, and $P_{i,t}$ the close price of stock *i* on day t. We further define $MCP^{*}{}_{i,j,t}$ as the natural log of (1+ $MCP_{i,j,t}$).

We estimate Fama-MacBeth regressions of realized correlations of each stock pair on lagged *MCP**:

$$\rho_{i,j,t+1} = a + b * MCP^{*}{}_{i,j,t} + \sum_{k=1}^{n} b_k * CONTROL_{i,j,k} + \varepsilon_{i,j,t+1}\,, \qquad (4)$$

where $\rho_{i,j,t+1}$ is the product of the two stocks' excess returns on day *t*+1. Following Anton and Polk (2014), we also control for a host of variables that are known to be associated with stock return correlations: a dummy variable that equals 1 if the two stocks are from the same industry, the absolute differences between the two stocks in firm size, book-to-market ratio, and momentum percentiles, and common analyst coverage. Moreover, as a placebo test, we randomly select 200,000 brokerage non-margin accounts, and compute similar connected measures on these, except that the leverage ratio is a constant one across all these accounts (*MCP_nm**).

Insert Table 3 about Here

Table 3 presents results of forecasting stock pairwise stock return correlations for the pooled sample between May and July, 2015, as well as for the shadow-financed sample between January and July, 2015. As shown in Column 1, the coefficient on MCP^{*} is 0.016 and a t-statistic of 3.37. Controlling for similarity in firm characteristics and common analyst coverage (shown in Column 2) has virtually no impact on this result. Column 3 further includes common ownership measures based on non-margin accounts. While MCP^{*} remains economically and statistically significant, the coefficient on MCP_nm^{*} is economically and statistically insignificant. This is suggestive that the effect we document is due to the contagion caused by levered trading.

In Columns 4 and 5, we repeat our analysis in Columns 1 and 2, but now focus solely on the shadow-financed subsample for the period of January to July, 2015. The regression coefficients are similar in magnitude to those reported in Columns 1 and 2 and are all statistically significant.

B. Forecasting Stock Returns with Connected Stock Returns

To the extent that investors scale up or down their holdings in response to the tightening or loosening of leverage constraints with a delay, the contagion effect we document above can lead to stock return predictability. To illustrate, imagine that stock A is initially hit by a negative shock, which then lowers the portfolio value of all investors holding stock A. If these investors start to sell other positions in their portfolios in the next day or week, then stock A's return is a leading indicator of other stocks with which is shares common ownership by levered investors.

To test this hypothesis, for each stock, we compute its connected portfolio return as the weighted average returns of all stocks with which it shares common holdings by margin investors. We then regress future stock returns on the connected portfolio return, along with other controls that are known to forecast stock returns:

$$RET_{i,t+1} = a + b * SCORE_{i,t} + \sum_{k=1}^{n} b_k * CONTROL_{i,k} + \varepsilon_{i,t+1}\,, \tag{5}$$

where $SCORE_{i,t}$ is the connected portfolio return for stock *i* on day *t*. The set of control variables is the same as that in Table 2.

Insert Table 4 about Here

As shown in Table 4, the connected portfolio return significantly and positively forecasts future stock returns. This holds even after controlling for the stock's own leverage and its own lagged returns.

4.3 Empirical Analysis based on a Network of Leveraged Common Holding

In this section, we construct a leverage network across all stocks in the Chinese market based on their common ownership by levered investors. In so doing, we can more directly examine the transmission mechanism of leverage-induced price impact in the entire cross-section of stocks.

A. Network Representation of Leveraged Common holdings

Our leverage network is defined by a symmetric adjacency matrix, denoted A, where each entry $a_{i,j}$ represents the level of "connectedness," as measured by equation (2), between stocks *i* and *j*. If there is no common ownership in the two stocks by levered investors, the connectedness measure is set to zero. Given the computational complexity, we focus on the constituents of the Zhongzheng 800 index, to form an 800*800 matrix.

It is important to note that through this network structure, shocks to one stock can be transmitted to stocks that are not directly linked to it. To illustrate, imagine stock A that is connected to stock B via common ownership, which is also linked to stock C. Negative shocks to stock A can trigger selling pressure on stock B, which is further transmitted to stock C, even though stocks A and C may not share any common investors. To capture this indirect relation, we use $(A)^n$ – that is, matrix A raised to the power of *n* – to measure how shocks may be transmitted from one stock to another going through *n* intermediary steps.

B. Network Centrality

A number of measures have been proposed in prior literature to quantify the importance of each node in a given network. These include degree, closeness, betweenness, and eigenvector centrality. Borgatti (2005) reviews these measures and compares their advantages and disadvantage based on their assumptions about how traffic flows in the network. Following Ahern (2015), we use the eigenvector centrality as our main measure of leverage network centrality.

Eigenvector centrality is defined as the principal eigenvector of the network's adjacency matrix (Bonacich, 1972). A node is more central if it is connected to other nodes that are themselves more central. Hence, if we define the eigenvector centrality of node *i* as c_i, then c_i is proportional to the sum of the c_j's for all other nodes $j \neq i$; $c_i = \frac{1}{\lambda}\sum_{j \in M(i)} c_j = \frac{1}{\lambda}\sum_{j=1}^{N} A_{ij c_j}$, where $M(i)$ is the set of nodes that are connected to node i and λ is a constant. In matrix notation, we have Ac = λ c. Thus, c is the principal eigenvector of the adjacency matrix.

The intuition behind eigenvector centrality is closely related to the stationary distribution. The Perron-Frobenius theorem stipulates that every Markov matrix has an eigenvector corresponding to the largest eigenvalue of the matrix, which represents the stable stationary state. Equivalently, this vector can be found by multiplying the transition matrix by itself infinite times. As long as the matrix has no absorbing states, then a non-trivial stationary distribution will arise in the limit. If we consider the normalized adjacency matrix as a Markov matrix, eigenvector centrality then represents the stationary distribution that would arise as shocks transition from one stock to another for an infinite number of times.

In sum, eigenvector centrality provides a measure of how important a node is in the network. It directly measures the strength of connectedness of a stock, considering the importance of the stocks to which it is connected. Equivalently, by tracing out all paths of a random shock in a network, eigenvector centrality measures the likelihood that a stock will receive a random shock that transmits across the network.

Insert Table 5 about Here

We start by analyzing the stock characteristics that are associated with the centrality measure. As shown in Table 5, stocks with larger market capitalization, higher turnover, and lower past returns tend to be more central in the leverage network structure.

C. Network Centrality, Stock Returns, and Stock Skewness.

Given that central stocks are likely to be hit by idiosyncratic shocks to any stocks in the network, and that negative shocks are more likely to be propagated through the network than positive shocks, we expect that stocks with a higher centrality score to have lower unconditional returns and higher crash risk (negative return skewness) compared to stocks with a lower centrality score.

To test this prediction, we conduct a Fama-MacBeth regression of individual stock returns on the stocks' eigenvector centrality measured in the previous day:

$$RET_{i,t+1} = a + b * CENTRAL_{i,t} + \sum_{k=1}^{n} b_k * CONTROL_{i,k} + \varepsilon_{i,t+1}\,, \tag{6}$$

where $CENTRAL_{i,t}$ is the standardized centrality measure for stock *i* measured in day t. We include the same set of control variables as in Table 2

Insert Table 6 about Here

Table 6 shows the regression results. As can be seen from Column 1, eigenvector centrality significantly and negatively forecasts future stock returns with a t-statistic of 2.62. This result is virtually unchanged after controlling for the stock's own past returns and its own leverage ratio.

Insert Table 7 about Here

Next, we analyze the relation between crash risk and eigenvector centrality by regressing future return skewness on our centrality measure:

$$SKEW_{i,t+1} = a + b * CENTRAL_{i,t} + \sum_{k=1}^{n} b_k * CONTROL_{i,k} + \varepsilon_{i,t+1}\,, \qquad (7)$$

where $SKEW_{i,t+1}$ is the average daily skewness of stock i in the five day between t+1 to t+5. Daily return skewness, in turn, is calculated based on stock returns over 10-minute intervals. The centrality measure and other control variables are identical to those in equation (6).

As shown in Table 7, eigenvector centrality significantly and negative forecast stock return skewness in the next 5 days, with a t-statistic of -2.35. Again, controlling for various firm characteristics (such as past returns and past leverage ratio) has little impact on this result.

5. Conclusion

Investors can lever up their positions by borrowing against the securities they hold. This practice subjects margin investors to the impact of funding conditions. A number of recent studies theoretically examine the interplay between funding conditions and asset prices. Testing these predictions, however, has been empirically challenging, as we do not directly observe investors' leverage ratios and stock holdings. In this paper, we tackle this challenge by taking advantage of unique account-level data from China that track hundreds of thousands of margin investors' borrowing and trading activities at a daily frequency.

Our main analysis covers a three-month period of May to July 2015, during which the Chinese stock market experienced a rollercoaster ride: the Shanghai Stock Exchange (SSE) Composite Index climbed 15% from the beginning of May to its peak at 5166.35 on June 12th, before crashing 30% by the end of July. Major financial media around the world have linked this incredible boom and bust in the stock market to the popularity of, and subsequent government crackdown on, margin trading in China.

We show that idiosyncratic shocks in the market can cause contagion across assets when these assets are "connected" through common holdings by margin investors. In particular, the returns of one security strongly and positively forecast the returns of other securities with which it shares a common margin investor base. Further, using a network-based approach, we show that stocks that are connected to more other stocks through common holdings by margin investors (i.e., that are more central to the leverage network) tend to have lower returns and higher crash risk going forward. Finally, we provide evidence that average connectedness of the leverage network help forecast future market returns in our sample.

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Figure 1. Shanghai Composite Index and Outstanding Brokerage-financed Margin Debts in the A-Share Stock Market

This figure depicts the Shanghai Composite Index (bars, left scale) and the total outstanding brokerage-financed margin debts (line, right scale, in billion yuan). The time period is from October1st, 2014 to August 31st, 2015.

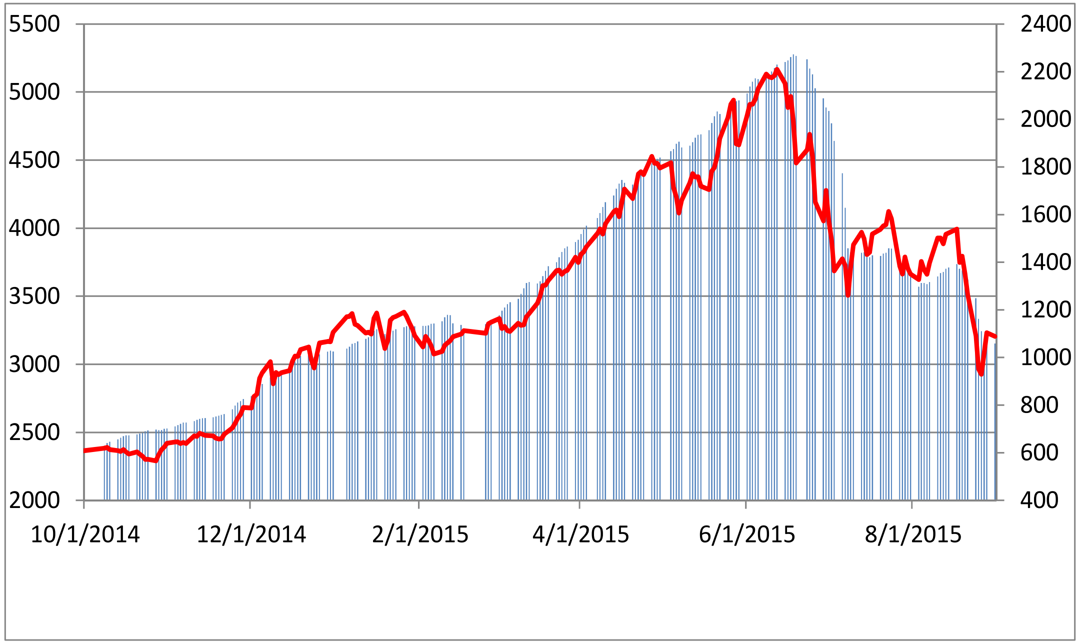

Figure 2. Wealth-Weighted Average Leverage Ratios for both Brokerage-Financed and Shadow-Financed Margin Markets

This figure depicts the time-series of the account wealth-weighted average leverage ratios for both the Brokerage-Financed and Shadow-Financed margin accounts in our sample. The time period is from October 1st 2014, to July 31st, 2015. There are 177,571 Brokerage-Financed margin accounts and 200,351 Shadow-Financed margin accounts. Leverage ratio is defined as the ratio of end-of-the-day total wealth to outstanding equity for each account. Daily market leverage ratio is weighted using each account's end-of-the-day total wealth, which includes Yuan holdings of both equity and cash.

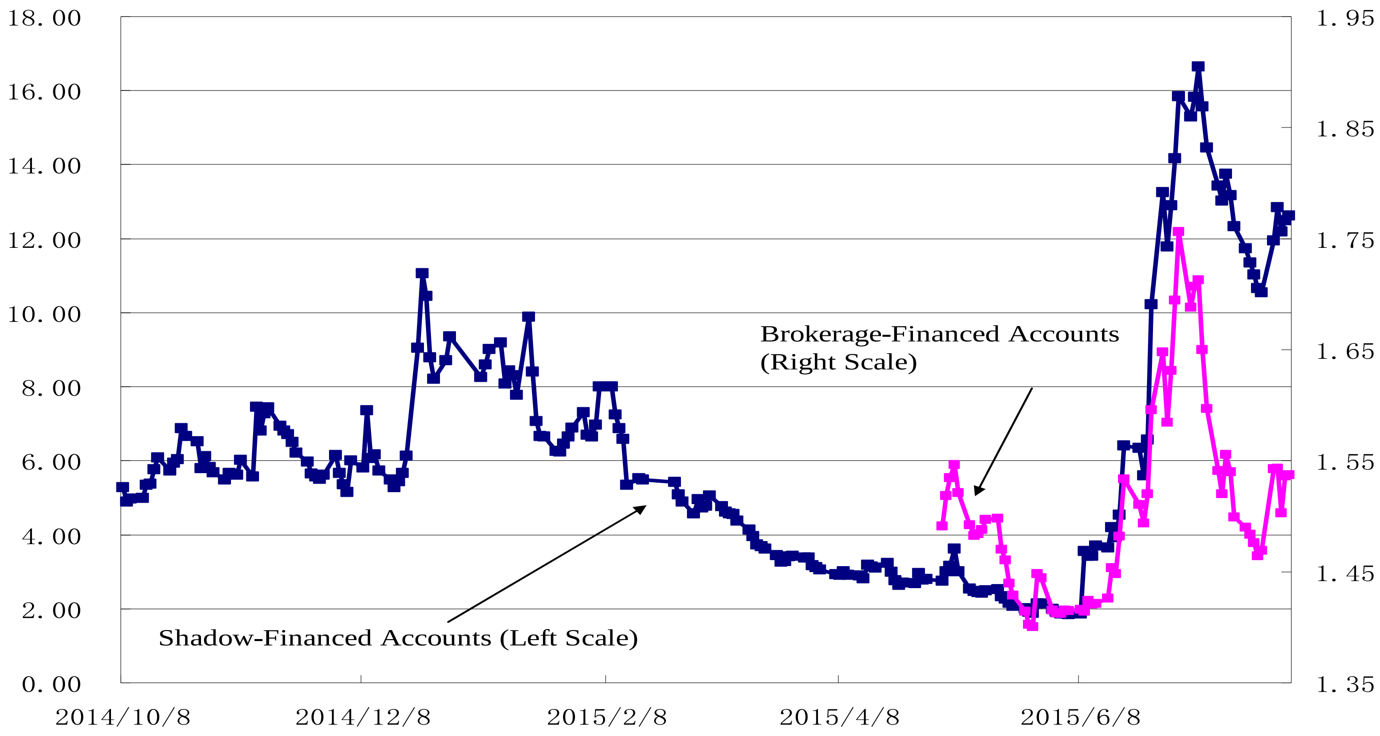

Table 1. Summary Statistics

This table reports statistics of the main sample used in this paper. The time period is from May 1st to July 30th 2015 Panel A reports the number of margin trading accounts and non-margin trading accounts in our brokerage-finance data and the number of margin accounts in our shadow-finance data. Panel A also reports the average daily total margin debt and total wealth for the brokerage-financed margin and non-margin accounts and the shadow-financed margin accounts. As a comparison, Panel A reports the average daily market capitalization of all A shares and tradable shares in the Chinese market. Panel B reports the mean (Mean) and median (Median) characteristics for investor accounts and stocks in our sample. These characteristic variables include the daily trading volume in shares (TRDVOL), daily dollar trading volume in Yuan ($TRDVOL), daily number of order submissions (ORDER), the number of stock shares held in each account at the end of each trading day (HOLD), and the leverage ratio that equals to the ratio of the total asset to the total debt of each account (LEVERAGE).

	Brokerage-financed margin accounts		Brokerage - financed non-margin accounts		Shadow - financed margin accounts	
	Panel A: Sample and Market Data Summary					
# of Accounts	177,571		5,726,109		200,351	
Average Total DEBT (billion Yuan)	136.4		0		44.5	
Average Total Wealth (billion Yuan)	355.0				91.6	
Average Total MCAP (billion Yuan)	57, 229					
Average Tradable MCAP (billion Yuan)	46, 060					
	Panel B: Accounts Characteristics					
	Mean	Median	Mean	Median	Mean	Median
TRDVOL(1,000 shares)	126.39	14.90	11.07	1.80	28.40	5.30
$ TRDVOL (10,000 Yuan)	208.14	27.30	18.34	2.80	51.05	10.23
# SUBMISSIONS	15.18	6.00	6.08	3.00	6.76	4.00
HOLD (1,000 shares)	362.18	62.80	29.02	3.80	64.09	7.60
LEVERAGE	1.60	1.53	1	1	9.23	4.57

Table 2
Stock Returns and Leverage Ratios

This table examines the relationship between the stock return and the stock leverage ratio at the end of the previous trading day. Our sample period covers the three month period from May to July, 2015. Panel A presents the results from a portfolio double sort. We first sort all trading days during our three month period into terciles based on market returns in the previous trading day. Then on each trading day, we sort all stocks into terciles based on their leverage ratios at the end of the previous trading day. We use Shanghai Composite Index return to measure the market return, and use two different methods to compute the stock leverage ratios. The first method uses only public data and computes the leverage as the ratio between the market capitalization and the absolute difference between the market capitalization and the margin debt. In the second method, we pool all the brokerage-financed and shadow-financed margin accounts. We then use each account's holdings of each stock as the weight to compute the weighted-average leverage ratio for each individual stock. Finally, Panel A reports average daily portfolio returns (both equal-weighted and value-weighted).Panel B regresses the stock's daily return on day t+1 to the stock's leverage ratio at the end of day t and other control variables for both the best and worst market return tertiles (boom and bust). The regression equation is as follows:

$$RET_{i,t+1} = \alpha_t + \beta_l LEVERAGE_{i,t} + \beta_c CONTROL_{i,t} + \varepsilon_{i,t+1}$$

$RET_{i,t+1}$ is the return of stock i on day t+1. $LEVERAGE_{i,t}$ is the leverage ratio for stock i at the end of day t. We use both public data and account-level data to compute the leverage ratios for each stock. $CONTROL_{i,t}$ is a vector of control variables of stock *i* day *t*. We include several control variables in the regressions. The control variables include the stock daily return at t (RET), total market capitalization for stock i at the end of previous month (MCAP), the cumulative daily stock return for the previous month (MOMENTUM), Average turnover ratios for the previous month (TURN), and standard deviation of the residual from regressing daily stock returns on Fama-French three factors during the 60 days prior to the end of the previous month (IDVOL). We report regression coefficients and associated *t*-values using Fama-Macbeth regressions. The *t*-values are reported in parentheses below the coefficients. The robust standard errors are adjusted for clustering first by each individual stock, and then by each trading day. ***, **, and * correspond to statistical significance at the 1%, 5%, and 10% levels, respectively.

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Panel A: Stock Returns for Double Sorted Portfolios, May to July, 2015

	Equally Weighted Return			MCAP Weighted Return		
	1 (Low Leverage)	2	3 (High Leverage)	1 (Low Leverage)	2	3 (High Leverage)
Public Market Data						
1 (Low Market Return)	-0.008	-0.015	-0.020	-0.001	-0.010	-0.014
2	0.001	0.0005	0.0015	-0.002	0.002	0.006
3 (High Market Return)	0.005	0.008	0.004	0.004	0.008	0.006
Account-Level Data						
1 (Low Market Return)	-0.003	-0.014	-0.025	-0.007	-0.014	-0.014
2	0.005	0.004	-0.003	0.0003	-0.006	-0.01
3 (High Market Return)	0.019	0.007	-0.007	0.012	0.001	-0.006

Panel B: Regression of Stock Return on Leverage Ratios and other Control Variables, May to July, 2015

	LEVERAGE	RET	MCAP	TURN	MOMENTUM	IDVOL
Market Condition: Boom						
Public Market Data	0.007	0.31**	-0.001	-0.015	0.002	-0.060
	(0.40)	(2.83)	(-0.50)	(-0.86)	(1.46)	(-0.37)
Account-Level Data	-0.0005	0.336***	-0.002	-0.039*	0.001**	-0.107
	(-0.88)	(3.04)	(-1.68)	(-1.92)	(2.57)	(-0.57)
Market Condition: Bust						
Public Market Data	-0.015***	0.20***	0.003	-0.069**	-0.006***	-0.047
	(-3.32)	(3.57)	(1.58)	(-2.78)	(-4.51)	(-0.23)
Account-Level Data	-0.007**	0.258***	0.004	0.021	-0.002*	-0.159
	(-2.06)	(6.32)	(1.67)	(0.88)	(-2.01)	(-0.84)

Table III
Stock Price Co-Movements and Connectedness Measures

We regress the daily measure of price co-movement between stock *i* and stock *j* that are commonly held by a group of margin investors on the connectedness measures, as well as other control variables. The regression equation is as follows:

$$\rho_{i,j,t+1} = \alpha_t + \beta_n MCP^{*}_{i,j,t} + +\beta_C CONTROL_{i,j,t} + \varepsilon_{i,t+1}$$

$\rho_{i,j,t+1}$ is the correlation variable, which equals to the product of daily excess returns on day *t+1* of the stock *i* and stock *j* that have been commonly held . $MCP^{*}_{i,j,t}$ is the variable that measures the connectedness between stock *i* and *j* at day *t*. *MCP* is calculated as the sum of the product of each investor's leverage ratio and the Yuan holding of the two connected stocks by the investor, divided by the sum of market capitalizations of the two connected stocks over 1000. We then take the natural log of (*1+MCP*). $CONTROL_{i,j,t}$ is a vector of control variables of the stock pair on day *t*. It includes the negative of the absolute difference in size, book-to-market ratio, and momentum percentile rankings between the two stocks ($SAMESIZE_{i,j,t}$, $SAMEBM_{i,j,t}$, and $SAMEMOM_{i,j,t}$). *NUMSIC* is equal to 1 if stock *i* and stock *j* are from the same industry as classified by the CSRC, and 0 otherwise. $A_{i,j,t}$ is number of different analysts who cover both stock *i* and stock *j* from day *t-250* to day *t-1*. We also include controls based on the normalized rank-transform of the percentile market capitalization of the two stocks, $SIZE_1$ and $SIZE_2$, as well as the interaction between them. Finally, we include the connected measures using the common holdings of 200,000 randomly selected brokerage non-margin investors, MCP_nm . For the non-margin investors, we assume the leverage to be 1. We report regression results for two sub-samples. The first subsample contains all the brokerage-financed and shadow-financed accounts from May to July, 2015. The second subsample contains the shadow-financed margin accounts from January to July, 2015.

We report regression coefficients and associated *t*-values using Fama-Macbeth regressions. The *t*-values are reported in parentheses below the coefficients. The robust standard errors are adjusted for clustering first by each individual stock, and then by each trading day. ***, **, and * correspond to statistical significance at the 1%, 5%, and 10% levels, respectively.

Dependent Variable: Product of Excess Returns of the Two Paired Stocks

	Brokerage-Financed + Shadow-Financed Margin Accounts: May-July, 2015			Shadow-Financed Margin Accounts: January - July, 2015	
	(1)	(2)	(3)	(4)	(5)
*MCP**	0.016***	0.018***	0.018***	0.027***	0.011***
	(3.37)	(3.02)	(3.01)	(6.79)	(6.79)
A		0.0002	0.0002		0.0003***
		(1.32)	(1.31)		(5.78)
SAMESIZE		0.007***	0.007***		0.006***
		(2.89)	(2.89)		(3.78)
SAMEBM		0.002***	0.002***		0.002***
		(4.31)	(4.31)		(5.67)
SAMEMOM		0.002***	0.002**		0.001***
		(3.62)	(3.61)		(3.61)
NUMSIC		0.012***	0.012***		0.015***
		(5.84)	(5.84)		(7.19)
SIZE1		0.002	0.002		0.002
		(1.47)	(1.48)		(3.13)
SIZE2		0.002	0.002		0.002
		(1.47)	(1.48)		(3.13)
*SIZE1*SIZE2*		-0.001**	-0.001**		-0.001
		(-2.56)	(-2.56)		(-3.78)
MCP_nm			0.007		
			(1.59)		

Table IV
Predicting the Returns of Connected Stocks

We regress the next-day stock returns on their connected portfolio returns today, as well as other control variables. The regression equation is as follows:

$$RET_{i,t+1} = \alpha_t + \beta_n SCORE_{i,t} + \beta_C CONTROL_{i,t} + \varepsilon_{i,t+1}$$

$RET_{i,t+1}$ is the return of stock *i* on day *t+1* . $SCORE_{i,t}$ is the day-t weighted average return on a portfolio of stocks that are connected to stock *i* through common holdings of margin traders (both brokerage-financed and shadow-financed). We use the connectedness measure in Table III as the weight. $CONTROL_{i,t}$ is a vector of control variables for stock *i* on day *t*. The control variables include the stock daily return at t ($RET_{i,t}$), total market capitalization for stock i at the end of previous month (*MCAP*), the cumulative daily stock return for the previous month (*MOMENTUM*), Average turnover ratios for the previous month (*TURN*), and standard deviation of the residual from regressing daily stock returns on Fama-French three factors during the 60 days prior to the end of the previous month (*IDVOL*). We also include stock leverage ratio at day *t* as the control variable ($LEVERAGE_{i,t}$). We compute the leverage ratio as the weighted average leverage ratio of the leverage ratio on each margin account that holds stock *i* on day *t*. We use each account's holding as the weight. The sample period is from May to July, 2015.

We report regression coefficients and associated *t*-values using Fama-Macbeth regressions. The *t*-values are reported in parentheses below the coefficients. The robust standard errors are adjusted for clustering first by each individual stock, and then by each trading day. ***, **, and * correspond to statistical significance at the 1%, 5%, and 10% levels, respectively.

Dependent Variable: Stock Return at day *t+1*, May to July, 2015

	(1)	(2)	(3)	(4)	(5)	(6)
SCORE	0.75***	0.28**	0.28**	0.30**	0.27**	0.20***
	(3.44)	(2.17)	(2.39)	(2.37)	(2.15)	(2.25)
LEVERAGE		0.0004	0.0004	0.0005	0.0005	0.001
		(0.29)	(0.29)	(0.41)	(0.37)	(0.59)
RET		0.21***	0.20***	0.20***	0.20***	0.19***
		(5.37)	(5.46)	(5.33)	(5.38)	(5.27)
IDVOL			-0.19			-0.08
			(-1.60)			(-0.92)
MOMENTUM				-0.002*		-0.001
				(-1.69)		(-1.17)
TURN					-0.04**	-0.035***
					(-2.28)	(-3.12)
MCAP						-0.001
						(-1.10)

Table V
Characteristics of the Stock Centrality Measure

We regress the daily stock centrality measure on the other stock characteristic variables. The regression equation is as follows:

$$CENTRAL_{i,t} = \alpha_t + \beta_C CONTROL_{i,t} + \varepsilon_{i,t}$$

$CENTRAL_{i,t}$ is the centrality measure of stock i on day t , taken from the eigenvector of the leverage network matrix based on the common holding structure of margin traders in our sample (both brokerage- and shadow-financed). To facilitate comparisons over time, the centrality measure is standardized in the cross-section. $CONTROL_{i,t}$ is a vector of control variables for stock i on day t. The control variables include the stock daily return at t ($RET_{i,t}$), total market capitalization for stock i at the end of previous month(*MCAP*), the cumulative daily stock return for the previous month (*MOMENTUM*), Average turnover ratios for the previous month (*TURN*), and standard deviation of the residual from regressing daily stock returns on Fama-French three factors during the 60 days prior to the end of the previous month (*IDVOL*). We report regression results for Public data sample, and the data sample containing both brokerage and shadow-financed accounts. We also include stock leverage ratio at day t as the control variable ($LEVERAGE_{i,t}$). We compute the leverage ratio as the weighted average leverage ratio of the leverage ratio on each margin account that holds stock i on day t. We use each account's holding as the weight. The sample period is from May to July, 2015.

We report regression coefficients and associated t-values using Fama-Macbeth regressions. The t-values are reported in parentheses below the coefficients. The robust standard errors are adjusted for clustering first by each individual stock, and then by each trading day. ***, **, and * correspond to statistical significance at the 1%, 5%, and 10% levels, respectively.

Dependent Variable: Stock Centrality Measure at day *t*, May to July, 2015

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
LEVERAGE	0.07 (0.75)						0.04 (0.53)
RET		0.01 (0.4)					-0.06 (-0.11)
MOMENTUM			-0.06 (-1.04)				-0.13*** (-3.12)
TURN				1.58** (2.24)			4.06*** (5.39)
IDVOL					-2.90 (-0.67)		2.42 (1.03)
MCAP						0.28*** (24.05)	0.30*** (26.31)

Table VI
Stock Centrality Measures and future returns

We regress next-day stock returns on the stock centrality measure today as well as other control variables. The regression equation is as follows:

$$RET_{i,t+1} = \alpha_t + \beta_n CENTRAL_{i,t} + \beta_C CONTROL_{i,t} + \varepsilon_{i,t+1}$$

$RET_{i,t+1}$ is the return of stock *i* on day *t+1* . $CENTRAL_{i,t}$ is the centrality measure of stock *i* on day *t* . $CONTROL_{i,t}$ is a vector of control variables for stock *i* on day *t*. The control variables include the stock daily return at t ($RET_{i,t}$), total market capitalization for stock i at the end of previous month(*MCAP*), the cumulative daily stock return for the previous month (*MOMENTUM*), Average turnover ratios for the previous month (*TURN*), and standard deviation of the residual from regressing daily stock returns on Fama-French three factors during the 60 days prior to the end of the previous month (*IDVOL*). We also include stock leverage ratio at day *t* as the control variable ($LEVERAGE_{i,t}$). We compute the leverage ratio as the weighted average leverage ratio of the leverage ratio on each margin account that holds stock *i* on day *t*. We use each account's holding as the weight. The sample period is from May to July, 2015.

We report regression coefficients and associated *t*-values using Fama-Macbeth regressions. The *t*-values are reported in parentheses below the coefficients. The robust standard errors are adjusted for clustering first by each individual stock, and then by each trading day. ***, **, and * correspond to statistical significance at the 1%, 5%, and 10% levels, respectively.

Dependent Variable: Stock Return on Day *t+1*, May to July, 2015

	(1)	(2)	(3)	(4)	(5)	(6)
CENTRAL	-0.0006**	-0.0007**	-0.0008***	-0.0003**	-0.0007**	-0.0006*
	(-2.62)	(2.28)	(-2.73)	(-2.13)	(-2.17)	(-1.97)
LEVERAGE		0.0006	0.0003	0.0005	0.0005	0.0007
		(0.55)	(0.32)	(0.54)	(0.56)	(0.68)
DRET		0.216***	0.22***	0.20***	0.20***	0.20***
		(4.11)	(4.01)	(4.24)	(4.20)	(4.18)
IDVOL			-0.227	-0.229	-0.152	-0.113
			(1.48)	(-1.55)	(-1.32)	(-1.00)
MCAP				-0.001	-0.0007	-0.0008
				(-0.61)	(-0.48)	(-0.61)
MOMENTUM					-0.001	-0.001
					(-1.51)	(-1.59)
TURN						-0.033**
						(-2.37)

Table VII
Stock Centrality Measures and the Skewness of Future Returns

We regress the stock future return skewness on its centrality measure today, as well as other control variables. The regression equation is as follows:

$$SKEW_{i,t+1} = \alpha_t + \beta_n CENTRAL_{i,t} + \beta_C CONTROL_{i,t} + \varepsilon_{i,t+1}$$

$SKEW_{i,t+1}$ is the average daily stock skewness measure over the period from day *t+1* to day t+5. On each day, we construct 10-minute intraday return, and compute the daily skewness using these 10-minute returns. $CENTRAL_{i,t}$ is the centrality measure of stock *i* on day *t* . $CONTROL_{i,t}$ is a vector of control variables for stock *i* on day *t*. The control variables include the stock daily return at t ($RET_{i,t}$), total market capitalization for stock i at the end of previous month(*MCAP*), the cumulative daily stock return for the previous month (*MOMENTUM*), Average turnover ratios for the previous month (*TURN*), and standard deviation of the residual from regressing daily stock returns on Fama-French three factors during the 60 days prior to the end of the previous month (*IDVOL*). We also include stock leverage ratio at day *t* as the control variable ($LEVERAGE_{i,t}$). We compute the leverage ratio as the weighted average leverage ratio of the leverage ratio on each margin account that holds stock *i* on day *t*. We use each account's holding as the weight. The sample period is from May to July, 2015.

We report regression coefficients and associated *t*-values using Fama-Macbeth regressions. The *t*-values are reported in parentheses below the coefficients. The robust standard errors are adjusted for clustering first by each individual stock, and then by each trading day. ***, **, and * correspond to statistical significance at the 1%, 5%, and 10% levels, respectively.

Dependent Variable：Average Daily Stock Skewness of 10-minute Intraday Stock Return for Day *t+1* to day *t+5*, May to July, 2015

	(1)	(2)	(3)	(4)	(5)	(6)
	-0.048**	-0.052**	-0.052***	-0.033*	-0.035**	-0.038*
CENTRAL	(-2.35)	(-2.56)	(-2.63)	(-1.99)	(-1.97)	(-1.83)
		0.004	0.007	0.016	0.016	0.015
LEVER		(0.36)	(0.59)	(1.12)	(1.14)	(1.10)
		2.281***	2.048***	1.455***	1.493***	1.462***
DRET		(3.17)	(3.58)	(3.70)	(3.79)	(3.73)
			-3.908	-4.613	-1.967	-2.456
IDVOL			(-0.86)	(-1.01)	(-0.69)	(-0.69)
				-0.078**	-0.076**	-0.071**
MCAP				(-2.31)	(-2.17)	(-2.02)
					-0.047	-0.043
MOMENTUM					(-1.19)	(-1.18)
						0.0793
TURN						(0.55)